

Bench sheet: steps 1 and 2

Upload diagnostic firmware, prove the controller, then measure the string. Rev 0.4 amended.

Before you start

No cells connected for any of this. Use a current-limited bench supply, 200 mA limit, for VCC.
Do not connect the panel or the charger. R3 sits at the board end, not out at the string.
Duty does not limit current: only R3 and the pin do. Verify at 4.20 V and 100% duty.

1 Upload the diagnostic application firmware

The clock/fuse conversion is complete: internal 8 MHz, BOD 2.7 V, no bootloader. Do not repeat it.

Disconnect cells, panel, UART and LED string. Power from one verified target-compatible ISP source only.

Action	Command / pass result
Open firmware folder	cd SolarLights_Lite/firmware/lite_controller
Check the target	avrdude -c usbasp -p m328p -v -> signature 0x1e950f
Compile diagnostics	pio run -e pro8_debug -> SUCCESS
Upload and verify	external current avrdude; never pio -t upload

Disconnect all ISP wires before normal power returns. USB-UART is monitoring-only: target TX -> adapter RX,
OUT-/GND -> adapter GND; leave adapter VO/VCC, TX, DTR and CTS open. Monitor at 9600 baud.

Full clean-Mac setup, Arduino-as-ISP fallback and troubleshooting: firmware/lite_controller/PROGRAMMING.md

2 Check the diagnostic firmware on a current-limited bench supply

Use the debug environment above. It enables serial but keeps the real timing scale (TIME_SCALE = 1).

Check	Expect
Startup after reset	One-time field guide, then rate-limited named diagnostic fields
VCC at 3.80 V	vdd within 0.05 V of the meter (set BANDGAP_V = 1.10 x Vmeter / Vreported)
D9 PWM frequency	1.9 to 2.0 kHz, confirming the 8 MHz clock
Pull D2 to GND	cause=BUTTON_TEST, output=100%, button timer counts down
Cover / uncover LDR	DUSK / DAWN confirmation reaches 300 s before mode changes

3 Gate 5: string current with R3 = 47 ohm (settles 47 vs 56 ohm)

DMM in series with the string positive. Lights at 100% duty (LDR covered). Record what you read:

Supply	Expect	Measured	R3 temp	Notes
3.40 V	11 mA	_____	_____	_____
3.70 V	15 mA	_____	_____	_____
4.20 V	21 mA	_____	_____	_____

Decision rule

Over 22 mA at 4.20 V: fit 56 ohm and repeat the three readings (expect 18 / 13 / 9 mA).
Within tolerance but too dim from the street at 3 am: keep 47 ohm and accept the 21 mA peak,
or document the measurements before deliberately changing the commanded duty in firmware.

Also confirm: R3 and the D9 pin stay at ambient; the string is dark with M2 unplugged and during reset.

For sleep-current and deployment, build pio run -e pro8 then flash its .hex with external avrdude.